A.1 Project registration form

Mukesh Patel School of Technology Management & Engineering

SVKM's NMIMS (Deemed-to-be University), Indore Campus

Project registration form

Department: Artificial Intelligence and Data Science

Program: B.Tech in Artificial Intelligence and Data Science

Semester: _____

Project Team details

S No	SAP ID	Roll No	Name	Signature
1	70562300097	S024	Yash Ramchandani	
2	70562200088	S019	Simar Singh Khanuja	
3	-	-	-	-

Three broad domains in which the project team intends to work

S No	Domain	Project idea in brief
1	Generative AI: large language models and LLM agents	A memory layer for long-running LLM agents that decides what to keep, compress, forget and recall, so long conversations fit a fixed token budget (MemGate).
2		
3		

A.2 Topic approval form

Mukesh Patel School of Technology Management & Engineering

SVKM's NMIMS (Deemed-to-be University), Indore Campus

Topic approval form

Department: Artificial Intelligence and Data Science

Program: B.Tech in Artificial Intelligence and Data Science

Semester: VII

Project Team details

S No	SAP ID	Roll No	Name	Signature
1	70562300097	S024	Yash Ramchandani	
2	70562200088	S019	Simar Singh Khanuja	
3	-	-	-	-

Type of Project (Tick any one)

Type of project	
Application	
Product	
Research	✓

Project details

Project Title	MemGate: An Adaptive Memory Layer for Long-Running LLM Agents
Project Objectives	1. Build an adaptive memory decision engine that scores every message and decides whether to keep it verbatim, compress it, forget it, or recall it later, so that the prompt stays within a fixed token budget. 2. Build a three-tier memory store (recent turns, compressed gists, long-term facts with embeddings) held to a fixed storage budget, with retrieval to bring dropped information back when it is relevant. 3. Evaluate the memory policies on long multi-session conversation benchmarks (LoCoMo and others), changing one factor at a time, and report the trade-off between memory budget and answer quality with proper statistics.
Domain	Generative AI / Natural Language Processing: memory and context management for LLM agents
Motivation	LLM agents re-send their whole history on every turn, so cost and latency grow with the conversation and accuracy falls well before the context limit. Sliding windows forget the oldest messages whatever they contain, and summaries lose names, numbers and dates. On LongMemEval, oracle evidence gives about 92% accuracy against about 58% when the model selects its own: a 34-point gap caused by memory selection, not reasoning. Existing memory systems are evaluated end to end, so the value of each memory decision is unknown.
Expected outcomes	A working memory layer (Python library) that sits in front of any LLM; a measurement harness and results on public benchmarks with confidence intervals; a learned scorer for message importance; a chat application that shows every keep, compress and forget decision; a research paper.

Latest References

S No	Publication (Author, “Title”, Journal/Book/Conference, Date, Page, Volume, year)
1	C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica and J. E. Gonzalez, "MemGPT: Towards LLMs as Operating Systems," arXiv preprint arXiv:2310.08560, 2023.
2	P. Chhikara, D. Khant, S. Aryan, T. Singh and D. Yadav, "Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory," arXiv preprint arXiv:2504.19413, 2025.
3	W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan and Y. Zhang, "A-MEM: Agentic Memory for LLM Agents," Advances in Neural Information Processing Systems (NeurIPS), 2025.
4	A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri and Y. Fang, "Evaluating Very Long-Term Conversational Memory of LLM Agents," Proc. Annual Meeting of the Association for Computational Linguistics (ACL), 2024.
5	D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang and D. Yu, "LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory," arXiv preprint arXiv:2410.10813, 2024.

Status of the project after Review 1 Comments/Remarks	
Name and Signature of the guide	Dr. Praveen Modi, Assistant Professor
Name and signature of panel members for review 1	
Signature of HoD	
If any change in project title after review 1 presentation as per comments received from panel and areas of concern to be noted here	

A.3 Evaluation sheet for Review 1: Topic approval and feasibility

Mukesh Patel School of Technology Management & Engineering

SVKM's NMIMS (Deemed-to-be University), Indore Campus

Evaluation sheet for Review 1: Topic approval and feasibility

S No	SAP ID	Roll No	Student Name	Identification of Problem statement, Purpose and need of the project (5 marks)	Feasibility of the implementation (5 marks)	Total (10 marks)
1	70562300097	S024	Yash Ramchandani			
2	70562200088	S019	Simar Singh Khanuja			

Project: MemGate: An Adaptive Memory Layer for Long-Running LLM Agents

Name & Signature of the Project mentor (with Date)

Dr. Praveen Modi, Assistant Professor

Signature: _____ Date: _____

Name & Signature of the panel members (with Date)

Signature: _____ Date: _____

Signature: _____ Date: _____

Signature: _____ Date: _____

A.4 Evaluation sheet for Review 2: Midterm review

Mukesh Patel School of Technology Management & Engineering

SVKM's NMIMS (Deemed-to-be University), Indore Campus

Evaluation sheet for Review 2: Midterm review

S No	SAP ID	Roll No	Student Name	Literature Review / Market Survey (5 marks)	Clarity of the problem statement (5 marks)	Design / System level representation (5 marks)	Implementation / Results (5 marks)	Interim Report (5 marks)	Total (25 marks)
1	70562300097	S024	Yash Ramchandani						
2	70562200088	S019	Simar Singh Khanuja						

Project: MemGate: An Adaptive Memory Layer for Long-Running LLM Agents

Name & Signature of the Project mentor (with Date)

Dr. Praveen Modi, Assistant Professor

Signature: _____ Date: _____

Name & Signature of the panel members (with Date)

Signature: _____ Date: _____

Signature: _____ Date: _____

Signature: _____ Date: _____

A.5 Evaluation sheet for Review 3: External Panel Member Evaluation

Mukesh Patel School of Technology Management & Engineering

SVKM's NMIMS (Deemed-to-be University), Indore Campus

Evaluation sheet for Review 3: Internal and External Panel Member Evaluation

S No	SAP ID	Roll No	Student Name	Literature review and clarity of the problem statement (5 marks)	Design / System level representation (5 marks)	Implementation and results (5 marks)	Report (5 marks)	Presentation (5 marks)	Viva (5 marks)	Total (30 marks)
1	70562300097	S024	Yash Ramchandani							
2	70562200088	S019	Simar Singh Khanuja							

Project: MemGate: An Adaptive Memory Layer for Long-Running LLM Agents

Name & Signature of the Internal mentor (with Date)

Dr. Praveen Modi, Assistant Professor

Signature: _____ Date: _____

Name & Signature of the External Panel Member (with Date)

Signature: _____ Date: _____

Signature: _____ Date: _____

A.6 Evaluation sheet for Research paper

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Mukesh Patel School of Technology Management & Engineering

SVKM's NMIMS (Deemed-to-be University), Indore Campus

Evaluation sheet for Research paper

S No	SAP ID	Roll No	Student Name	Marks (10 marks)
1	70562300097	S024	Yash Ramchandani	
2	70562200088	S019	Simar Singh Khanuja	

Paper: "Selection Beats Compression: Memory Decision Policies for LLM Agents Under a Bounded Store", Yash Ramchandani, Simar Singh Khanuja and Dr. Praveen Modi.

Status (submitted / accepted / published, with venue and proof attached):

_____

Name & Signature of the Project mentor (with Date)

Dr. Praveen Modi, Assistant Professor

Signature: _____ Date: _____